

一人食纪要：家庭微波烹饪中的风味探索与水溶性维生素保留

景文 (Jingwen)

独立作者

通讯邮箱: goodmanjingwenzhou@gmail.com

摘要 本文记录并分享一种简单、可重复的一人食烹饪方法：将食材、容器、盐与微波炉作为全部所需工具，通过分层放置食材、撒盐、微波加热等步骤完成一餐。文章以一次完整操作（含各环节计时）为案例，结合营养化学的基本原理，讨论水溶性维生素（主要为维生素C与叶酸）在加热过程中的两条流失路径——氧化分解与溶出——及其触发条件、必要条件与相互作用关系，并据此提出在保证食物安全熟透的前提下，尽量减少水溶性维生素流失的实践建议。文末附有常见高维生素C、高叶酸食物的营养成分对照表及每日推荐摄入量参考，供读者在日常烹饪中参考使用。

关键词：微波烹饪；水溶性维生素；氧化降解；溶出；一人食；家庭营养

1. 引言

我想跟你分享我是怎么做饭的，同时我想教会你这种做饭的方法。对我来说，这样做出的一餐吃起来感觉很好。大部分时候一天一餐就足够了，除非那天做了强度很高的运动，因为这样做出来的饭很管饱。而且它能帮助我的身心变得更加美丽强大。我会选择在早上接近中午的时间吃。

这么棒的饭做出来会不会很麻烦呢？为了完成这次分享，我选择在今天的下午做一顿饭，同时做个计时。

我之所以喜爱这种做饭方式的另一个重要原因是，它简单，不容易出错，还耐玩。

2. 材料与amp;方法

2.1. 所需工具

这种做饭方式需要准备：食材、盛装食材的容器、盐、微波炉。

2.2. 操作步骤

这种做饭方式的基本法则：

1. 购买、储存、做前洗切（并解冻）食材。
2. 把食材放进容器，撒盐。
3. 把容器放进微波炉。
4. 让微波炉运作。
5. 把容器从微波炉拿出来，当心烫。
6. 吃。
7. 洗干净容器并擦干净微波炉。

我可以通过每一次做出来的饭的味道、口感和样子，更加生动地了解不同食物的本性。

我用到的调料是盐，不同的食物搭配在一起，最终在碗里收紧的汁水，味道是不同的。我发现，加尖椒与不加尖椒，味道差异很明显。番茄也是，我喜欢有番茄。还有，在碗里，哪个食材放在最下面，哪个放在中间，哪个放在上面；食物切成大块还是小块，或者索性不切；容器选择玻璃的，还是陶瓷的。这些都可以慢慢探索。

2.3. 本次实验所用食材

食材依次为：藕、山药、昨天剩下的半个杏鲍菇、尖椒、昨天剩下的冬瓜和西兰花、昨天剩下的芦笋根部（尖部已于前一日食用）；以及从冷冻室拿进冷藏室解冻好的肉，约200g牛腱和50g羊肉。

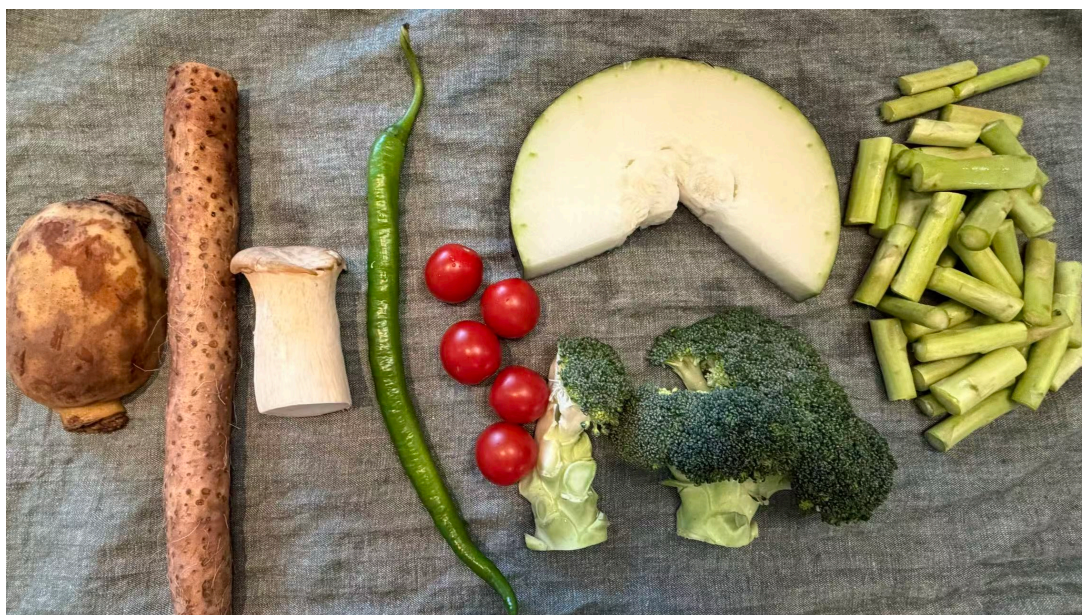


图 1: 食材从左往右依次是: 藕、山药、昨天剩下的半个杏鲍菇、尖椒、昨天剩下的冬瓜和西兰花、昨天剩下的芦笋根部（尖部被我昨天吃掉了）

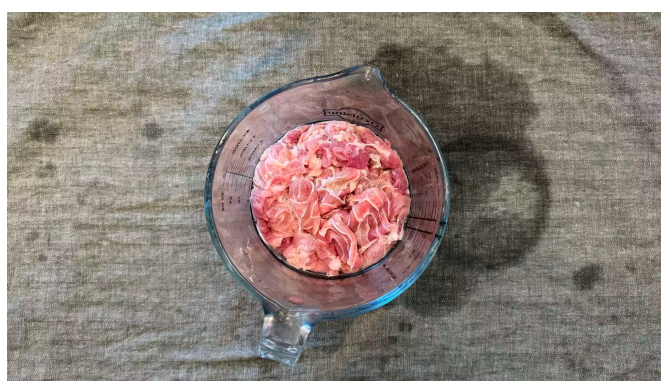


图 2: 从冷冻室拿进冷藏室解冻好的肉，约200g牛腱和50g羊肉。

2.4. 处理与计时

削好藕和山药的皮，并切好食材，用时6分21.00秒。把食物放进容器，撒点儿盐，用时5分32.07秒。

从下往上的食物依次是山药和藕、杏鲍菇、肉、西兰花、冬瓜、小番茄。尖椒塞进碗壁的缝缝里。芦笋这次没用上，留待下次。

2.5. 加热与食品安全

最后进微波炉，加热时间取决于不同微波炉的功率以及个人喜欢的熟度，这需要读者自己实验。如果肉没有提前解冻，需要的时间就更长一些。务必确保肉彻底熟透、没



图 3: 把食物放进容器, 记得撒点儿盐。

有生腥味或血水——有时肉的边缘已经熟了, 中心却可能还没有熟透, 这一点在食用前应加以确认。

3. 结果与讨论

3.1. 一个故事: 关于加热与营养流失

做这顿饭的时候, 有人跟我讲, 加热时间太长了, 会有营养流失。之前我不知道这个常识, 现在我知道, 这句话在当下是对的。

营养流失的意思是水溶性维生素暴露在热和氧气环境里的氧化分解过程。水溶性维生素主要是维生素C和叶酸。

氧化分解的必要条件 = 氧气 + 维生素C分子 (干燥无水环境下也会发生, 但极慢) 氧化分解的加速条件 = 热 或 水 (作为反应介质, 大幅提高分子碰撞频率和反应速率)

3.2. 营养流失的两条路径

营养流失主要有两种路径, 如表1所示。

两者关系: 路径一始终在发生 (速率因环境而异); 路径二只在满足"外部水体"条件时才叠加出现, 一旦叠加, 会反过来放大路径一的速率。

路径	触发条件	是否必要	典型场景
路径一：氧化	氧气 + 维生素分子 (热/时间/湿度加速)	必要，随时发生	采摘后存放、切开暴露空气、干燥保存、微波炉加热（加热环境没有除空气水分外的水体的情况）
路径二：溶出	大量外部连续水体	非必要，特定场景才触发	水煮、焯水、长时间浸泡

表 1: 水溶性维生素流失的两条路径

3.3. 应用于本次烹饪

微波炉的原理是它发射的微波使食物中的水分子（极性分子）高速振动摩擦生热。我做的这顿饭，冬瓜、小番茄、西兰花这些含水量高的食物在加热过程中持续释放水分，加上解冻的肉带了不少水，加热时间又长，水溶性维生素的营养流失速率因此被显著提高。

如果煮一锅富含水溶性维生素的蔬菜汤，由于加了不少水，营养流失速率是很快的。所以对于水溶性维生素而言，选择水煮的方式时，尤其需要把控火候，以尽量减少食物加热过程的营养损耗。

当我吃这一顿饭时，第一次觉得我的西兰花如此乏味。

3.4. 令人欣慰的另一面

不过，令我开心的是，我知道了这些东西：绝大部分营养成分，例如蛋白质、矿物质、膳食纤维以及脂溶性维生素，都是非常耐操的营养。

下面（表2）列出了维生素C和叶酸含量都比较高的食物，在做饭时，可以特别照顾一下这些食材的加热方式和时长。

表 2: 维生素C与叶酸含量对照表（除特别标注外，均为每100克可食部分，生重）

食物	维生素C (mg)	叶酸 (μg)	备注
芥蓝	约62	约141	两者都突出，很值得优先选
西兰花	约89	约57--63	两者都不错，尤其维C很高

食物	维生素C (mg)	叶酸 (μg)	备注
抱子甘蓝	约85	约61	两者都较高
红甜椒	约128	约46	维C极高，叶酸中等
菠菜	约28	约194	叶酸之王，维C一般
芦笋（熟）	约7--8	约52	按同等重量比，其实不算突出
猕猴桃	约93	约25	维C高，叶酸一般
草莓	约59	约24	维C不错，叶酸一般
橙子	约53	约30	均衡但不突出
生菜（罗马生菜等）	约4--24	约73--136	叶酸不错，维C较低
牛油果	约10	约81	叶酸不错，维C一般
鸡肝（熟）	约17--20	约510--770	叶酸最集中来源，波动较大（与品种、部位有关）
猪肝（熟）	约15--23	约330左右	叶酸略低于鸡肝，仍是优质来源
黑豆/红豆（熟）	约0--1.5	约130--150	叶酸高，几乎没有维C
扁豆（熟）	约1.5	约180	叶酸很高
鸡蛋（1个大号，约50g）*	0	约22	*按"个"计算，非100g，无法与其他行直接横向比较

营养素	每日推荐量	大约相当于
维生素C	100毫克	约1个中等猕猴桃（93mg）；或约80克红甜椒（不到半个）；或约110克西兰花（约小半棵）
叶酸	400微克	约280克菠菜（两把左右）；或约650克芦笋（一大把多）；或约130克鸡肝（一小块）

表 3: 每日推荐摄入量参考（中国营养学会成人标准）

数据参考USDA食品成分数据库，实际含量会因品种、产地、成熟度、储存及烹饪方式产生波动。

4. 结论

像西兰花，我们吃的是花蕾和嫩茎，这种组织结构本身娇嫩。如果想要吃它，还想要它的营养，去了解这些食物是应该的。

最后，我非常喜欢这一顿饭。

编辑完这一段文字已经是做完这顿饭的第二天中午，今天我吃到的芦笋，特别香嫩。

致谢 感谢那位提醒我"加热时间太长会造成营养流失"的朋友，是这句话促成了本文对水溶性维生素流失机制的进一步梳理。

参考文献

- [1] U.S. Department of Agriculture, Agricultural Research Service. FoodData Central. <https://fdc.nal.usda.gov/>
- [2] 中国营养学会. 《中国居民膳食营养素参考摄入量》.

Notes from a Solo Kitchen: Flavor Exploration and Water-Soluble Vitamin Retention in Home Microwave Cooking

Jingwen

Independent Author

Correspondence: goodmanjingwenzhou@gmail.com

Abstract This paper documents and shares a simple, repeatable method for cooking a single-serving meal, requiring only ingredients, a container, salt, and a microwave. Using one fully timed cooking session as a case study, the paper draws on basic nutritional chemistry to discuss the two pathways by which water-soluble vitamins (primarily vitamin C and folate) are lost during heating – oxidative degradation and leaching – along with their triggering conditions, necessary conditions, and interaction. Based on this, practical recommendations are proposed for minimizing water-soluble vitamin loss while still ensuring food is thoroughly and safely cooked. The paper concludes with a reference table of common foods high in vitamin C and folate, alongside recommended daily intake benchmarks, for use in everyday cooking.

Keywords: microwave cooking; water-soluble vitamins; oxidative degradation; leaching; solo cooking; home nutrition

1. Introduction

I want to share how I cook, and teach you this method too. For me, a meal made this way feels really good to eat. Most days, one meal a day is enough – unless I’ve done high-intensity exercise that day, since this kind of meal is very filling. It also helps my mind and body grow more beautiful and strong. I usually eat it in the late morning, close to noon.

Is a meal this good actually a hassle to make? To put this together, I chose to cook a meal this afternoon and time myself while doing it.

Another big reason I love this way of cooking is that it's simple, hard to mess up, and endlessly playful.

2. Materials and Methods

2.1. Equipment Required

This method requires: ingredients, a container to hold them, salt, and a microwave.

2.2. Procedure

The basic rules of this method:

1. Buy, store, and – before cooking – wash and cut (and thaw) the ingredients.
2. Put the ingredients in the container, sprinkle with salt.
3. Put the container in the microwave.
4. Let the microwave run.
5. Take the container out of the microwave – watch out, it's hot.
6. Eat.
7. Wash the container and wipe down the microwave.

Through the taste, texture, and appearance of each meal I make, I get a more vivid understanding of the nature of different foods.

The only seasoning I use is salt. Different foods combined together produce different flavors in the juices that collect at the bottom of the bowl. I've noticed that adding hot peppers versus not adding them makes a very noticeable difference. Same with tomatoes – I like having tomato in there. There's also the question of which ingredient goes on the bottom, which in the middle, which on top; whether the food is cut into large pieces, small pieces, or left whole; and whether the container is glass or ceramic. All of this is something to slowly explore.

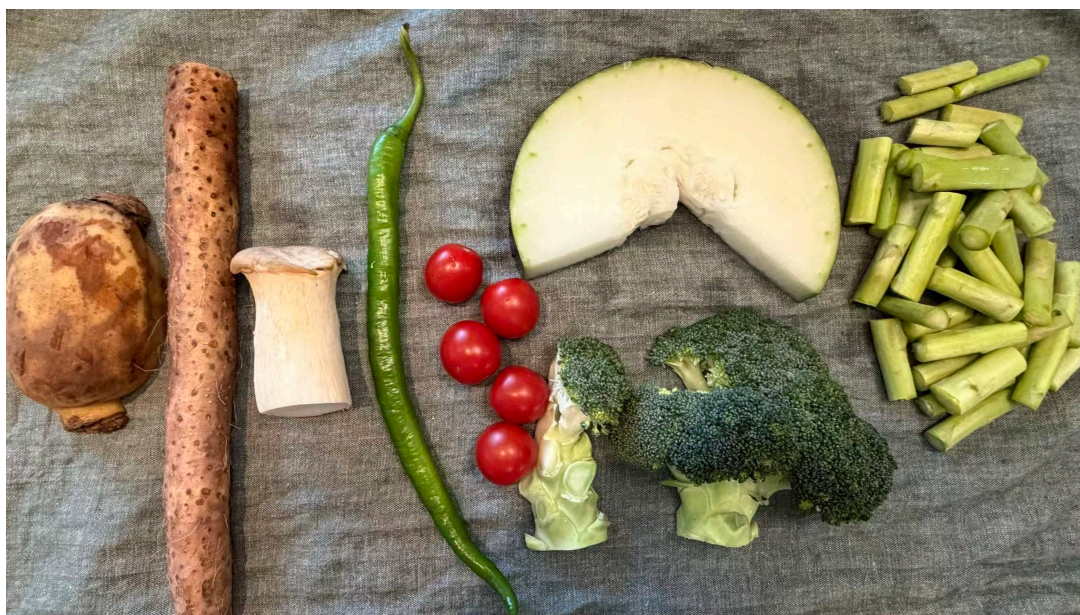


Figure 4: Ingredients from left to right: lotus root, Chinese yam, half a king oyster mushroom left over from the previous day, a hot pepper, winter melon and broccoli left over from the previous day, and the base of an asparagus stalk left over from the previous day (the tip having already been eaten).

2.3. Ingredients Used in This Session

Ingredients: lotus root, Chinese yam, half a king oyster mushroom left over from the previous day, a hot pepper, winter melon and broccoli left over from the previous day, and the base of an asparagus stalk left over from the previous day (the tip having already been eaten). In addition: meat taken out of the freezer and thawed in the refrigerator – about 200g beef shank and 50g lamb.

2.4. Preparation and Timing

Peeled the lotus root and yam, and cut up all the ingredients. Time: 6 minutes 21.00 seconds.

Put the food into the container, sprinkling a bit of salt. Time: 5 minutes 32.07 seconds.

From bottom to top: yam and lotus root, king oyster mushroom, meat, broccoli, winter melon, cherry tomatoes. The hot pepper was tucked into the gaps along the side of the bowl. The asparagus was not used this time – saved for next time.

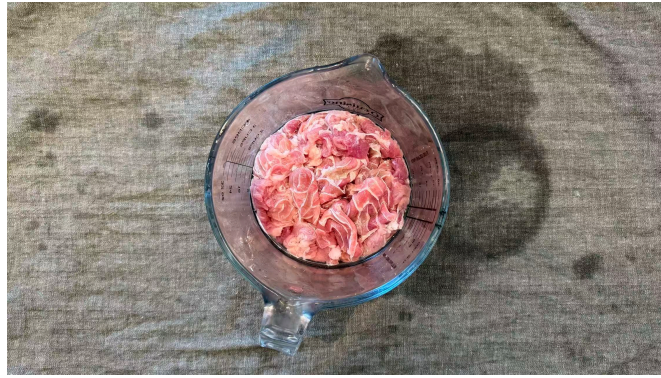


Figure 5: Meat taken out of the freezer and thawed in the refrigerator – about 200g beef shank and 50g lamb.

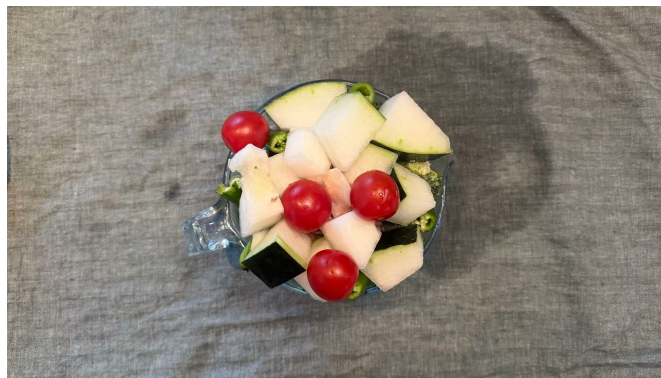


Figure 6: Put the food into the container, remembering to sprinkle on some salt.

2.5. Heating and Food Safety

Finally, into the microwave. The heating time depends on the particular microwave's power and how well-done one likes their food – this requires experimentation on the reader's part. If the meat hasn't been thawed ahead of time, it will need more time. It is important to make sure the meat is cooked all the way through, with no raw smell or blood – sometimes the outside is done while the center still isn't, and this should be checked before eating.

3. Results and Discussion

3.1. A Story: On Heating and Nutrient Loss

While making this meal, someone told me that heating food for too long causes nutrient loss. I didn't know this before. Now I know – and this statement is correct, as far as we currently understand it.

Nutrient loss here refers to the oxidative breakdown that water-soluble vitamins undergo when exposed to heat and oxygen. The water-soluble vitamins mainly at stake are vitamin C and folate.

Necessary condition for oxidative breakdown = oxygen + a vitamin C molecule
(this still happens in a dry, water-free environment, just extremely slowly)

Accelerating conditions = heat or water (water acts as a reaction medium, greatly increasing the frequency of molecular collisions and the reaction rate)

3.2. Two Pathways of Nutrient Loss

Nutrient loss mainly occurs through two pathways, summarized in Table 1.

Pathway	Trigger Condition	Necessary?	Typical Scenarios
Pathway 1: Oxidation	Oxygen + vitamin molecules (accelerated by heat/time/humidity)	Necessary – happens constantly	Storage after harvest, exposure to air after cutting, dry storage, microwave heating (in an environment without a water body beyond ambient air moisture)
Pathway 2: Leaching	A large, continuous external body of water	Not necessary – only triggered in specific scenarios	Boiling, blanching, prolonged soaking

Table 1: Two pathways of water-soluble vitamin loss

Relationship between the two: Pathway 1 is always happening (the rate varies with the environment). Pathway 2 only appears as an additional layer once the "external water body" condition is met – and once it does, it in turn accelerates Pathway 1.

3.3. Application to This Meal

The way a microwave works is that it emits microwaves causing water molecules (polar molecules) in the food to vibrate rapidly, generating heat through friction. In this meal, high-water-content foods such as winter melon, cherry tomatoes, and broccoli kept releasing water as they heated. On top of that, the thawed meat also carried a fair amount of water. With a long heating time,

the rate of water-soluble vitamin loss was significantly increased as a result.

If one boils a pot of vegetable soup rich in water-soluble vitamins, since a lot of water has been added, the rate of nutrient loss will be fast. So when it comes to water-soluble vitamins, if one chooses to boil food, it is especially important to control the heat and timing to minimize nutrient loss during cooking.

When I ate this meal, for the first time I found my broccoli tasted rather bland.

3.4. The Encouraging Side of the Story

Still, what makes me happy is what I learned along the way: the vast majority of nutrients – protein, minerals, dietary fiber, and fat-soluble vitamins – are all very robust and hold up well.

Table 2 below lists foods high in both vitamin C and folate, so that when cooking, one can pay special attention to heating method and duration for these in particular.

Table 2: Vitamin C and folate content (per 100g edible portion, raw weight, unless otherwise noted)

Food	Vitamin C (mg)	Folate (µg)	Notes
Chinese kale (gai lan)	~62	~141	Strong in both – worth prioritizing
Broccoli	~89	~57–63	Good in both, especially vitamin C
Brussels sprouts	~85	~61	Both fairly high
Red bell pepper	~128	~46	Very high vitamin C, moderate folate
Spinach	~28	~194	The king of folate; vitamin C average
Asparagus (cooked)	~7–8	~52	Not especially notable by equal weight
Kiwifruit	~93	~25	High vitamin C, average folate

Food	Vitamin C (mg)	Folate (µg)	Notes
Strawberries	~59	~24	Decent vitamin C, average folate
Orange	~53	~30	Balanced but not standout
Lettuce (romaine, etc.)	~4–24	~73–136	Decent folate, lower vitamin C
Avocado	~10	~81	Decent folate, average vitamin C
Chicken liver (cooked)	~17–20	~510–770	Most concentrated folate source; varies by breed/cut
Pork liver (cooked)	~15–23	~330 or so	Slightly less folate than chicken liver
Black beans / red beans (cooked)	~0–1.5	~130–150	High folate, almost no vitamin C
Lentils (cooked)	~1.5	~180	Very high folate
Egg (1 large, ~50g)*	0	~22	*Measured per egg, not per 100g – not directly comparable to other rows

Nutrient	Daily Recommended Amount	Roughly Equivalent To
Vitamin C	100 mg	About 1 medium kiwifruit (93mg); or about 80g red bell pepper (less than half a pepper); or about 110g broccoli (raw weight, a bit less than half a head)
Folate	400 µg	About 280g spinach (roughly two handfuls); or about 650g asparagus (a large handful or more); or about 130g chicken liver (a small piece)

Table 3: Reference daily intake (Chinese Nutrition Society, adult standard)

Data referenced from the USDA food composition database. Actual values will vary depending

on variety, origin, ripeness, storage, and cooking method.

4. Conclusion

With broccoli, what we eat is the flower buds and tender stem – a structure that is inherently delicate. If one wants to eat it and still get its nutrients, it is worth taking the time to understand foods like this.

In the end, I really enjoyed this meal.

I am finishing edits on this piece the day after making this meal, at noon. Today, the asparagus I ate was especially fragrant and tender.

Acknowledgments I thank the friend who reminded me that "heating for too long causes nutrient loss" – it was this remark that prompted the closer examination of water-soluble vitamin loss mechanisms presented in this paper.

References

- [1] U.S. Department of Agriculture, Agricultural Research Service. *FoodData Central*. <https://fdc.nal.usda.gov/>
- [2] Chinese Nutrition Society. *Chinese Dietary Reference Intakes*.